Machine Learning
Homework 4

Dec. 2025

Fill your Name
Fill your Student ID

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1. Here is a sample table. Please use the data in the table 1 to solve the decision tree-related problems.

Day	**Outlook**	**Temperature**	**Humidity**	**wind**	**Play Tennis**
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Hot	High	Strong	No

Table 1: Instances for building a decision tree.

   (a) Compute the Overall Entropy.

   (b) Compute the information gain and select the root node of the decision tree.

Ans:

(a)

(b)

2. Here are the results of three classifier ensembles, where C denotes the base classifier and C^* denotes the ensemble result. The three subplots correspond to three typical error distribution patterns of base classifiers.

Table 2: Ensemble 1

	t_1	t_2	t_3
C_1	✓	✓	×
C_2	×	✓	✓
C_3	✓	×	✓
C^*	✓	✓	✓

Table 3: Ensemble 2

	t_1	t_2	t_3
C_1	✓	✓	×
C_2	✓	✓	×
C_3	✓	✓	×
C^*	✓	✓	×

Table 4: Ensemble 3

	t_1	t_2	t_3
C_1	✓	×	×
C_2	×	✓	×
C_3	×	×	✓
C^*	×	×	×

(a) Please indicate which ensemble methods (Boosting / Random Forest / Bagging) can improve these error scenarios.

Ans:

(a)

3. Perform clustering analysis on the following two-dimensional data using the k-medoids algorithm. Assuming k=2, initialize the cluster centers using data points G and H. Calculate the similarity between any two points using Manhattan distance.

ID	A	B	C	D	E	F	G	H
feature 1	1.2	2.1	1.9	2.3	3.1	4.2	1.2	3.3
feature 2	1.1	2.4	3.1	4.2	4	4	1	2.2

Table 5: Instances for the k-medoids

(a) Provide the clustering results and key computational steps.

Ans:

(a)